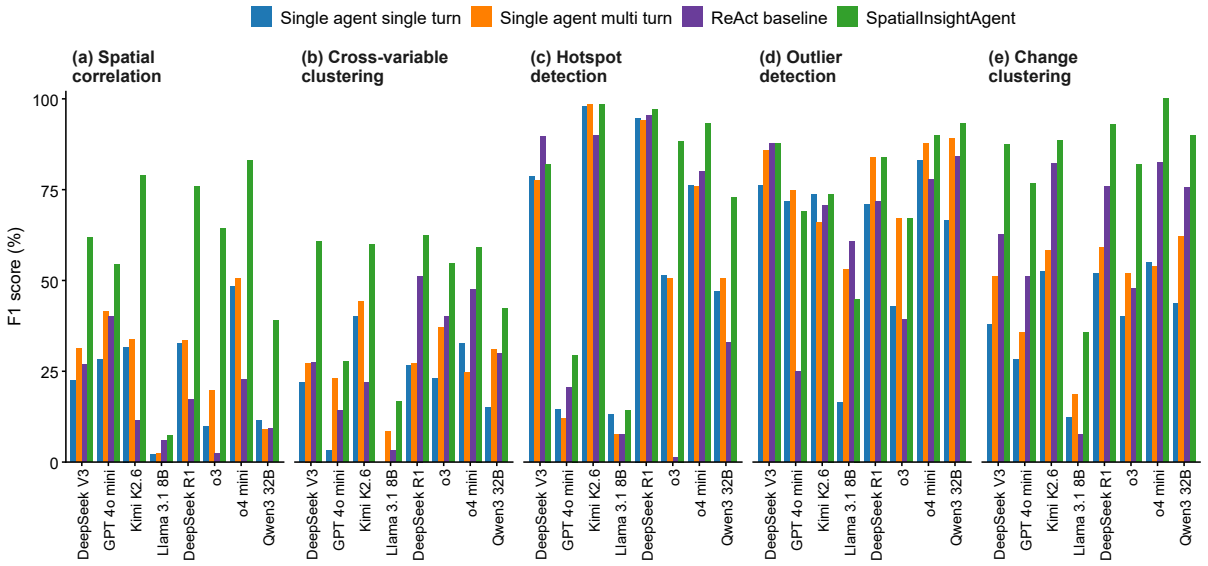
Single agent single turn
Single agent multi turn
ReAct baseline
SpatialInsightAgent
(a) Spatial correlation
(b) Cross-variable clustering
(c) Hotspot detection
(d) Outlier detection
(e) Change clustering
F1 score (%)
100
75
50
25
0
DeepSeek V3
GPT 4o mini
Kimi K2.6
Llama 3.1 8B
DeepSeek R1
o3
o4 mini
Qwen3 32B